

VOLUME 1

Overview and Safety

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-001
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Machine and retrofit scope

Machine: Mazak Vertical Quality Center VQC-20/40, serial 060231. Original control: Mazatrol M-2 / MELDAS. Shop supply is documented as 460 VAC three-phase into the retained machine transformer. The retrofit replaces the numerical-control and PLC interface functions while retaining the mechanical machine, Mitsubishi velocity-loop servo amplifiers, shared rectifier/capacitor bus, Tamagawa axis resolvers, FR-SX spindle drive, hydraulics, lubrication, coolant, and tool changer.

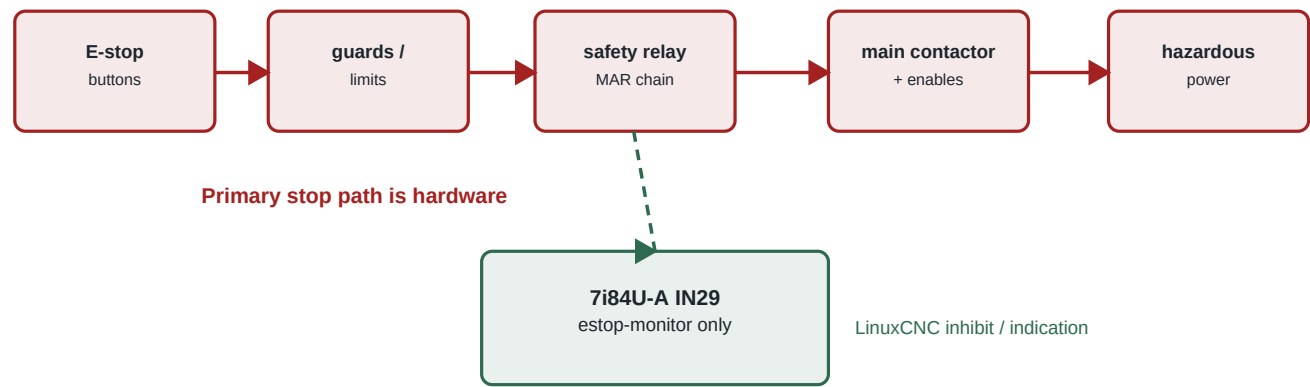
The retained equipment is obsolete and machine-specific. Adjacent-family Mitsubishi manuals are evidence for survey planning, not proof of the installed part or terminal behavior.

Function	Retained	Replaced / added
Motion power	MELDAS HD/TRA amplifiers, motors, tach loops, shared DC bus	LinuxCNC outer position loop and 7i49 analog commands
Feedback	Tamagawa TS2014N shaft resolvers	7i49 excitation and resolver-to-digital conversion
Spindle	Mitsubishi FR-SX, motor, orient detector, gearbox	7i49 AOUT3 plus 7i84U digital command/monitor
Field I/O	Switches, sensors, solenoids, relays	Two Mesa 7i84Us via 7i44
Safety	OEM MAR/hardwired chain intended to remain primary	LinuxCNC monitoring and supplemental inhibition only

2. Safety authority

LinuxCNC, hm2_eth, the 7i80HDT, 7i44, and 7i84U are not claimed as safety-rated equipment. The personnel-protection path must remove hazardous power through verified hardware without depending on the PC, Ethernet, FPGA, smart-serial, HAL, or the next servo cycle.

The checked-in estop-latch, watchdog, packet-error, drive-fault, and output-permit logic is a supplemental software inhibit. It cannot establish SIL, PL, stop category, guard integrity, or contactor redundancy.



The as-built hardware chain and fault-injection matrix are incomplete. No SIL/PL or stop-category claim is made.

Figure 1-1. Required authority split: hardware removes hazardous power; LinuxCNC monitors and inhibits.

3. Lockout, stored energy, and DC bus

- Open and lock the machine disconnect using the site's approved lockout/tagout procedure.
- Wait at least 10 minutes after power-off as required by the referenced Mitsubishi drive-family safety instructions.
- Do not trust elapsed time or a CHARGE lamp alone. Measure P(+) to N(-) with a properly rated meter and the exact installed-unit procedure.
- Proceed only below the service threshold specified by the identified rectifier/capacitor manual. The repository intentionally does not invent a universal voltage threshold.
- Repeat discharge characterization after rapid deceleration because regenerated energy can raise the bus above its at-rest condition.

- Support or crib the Z head before any brake, drive, or bus work.

SHARED BUS HAZARD

A single-axis fault does not isolate that amplifier from the common P/N bus. Bus discharge, regen, READY propagation, contactor behavior, and vertical-axis brake timing must be surveyed and measured as a group.

4. E-stop and guard-chain acceptance

- Draw the as-built chain with every button contact, guard, limit, safety relay, MAR relay, contactor, mirror contact, wire number, and removed load.
- Determine the achieved stopping category from hardware and measured stop behavior; do not assign Category 0 or 1 by assumption.
- Execute and sign the 18-case fault-injection matrix in docs/estop_safety_chain.md.
- Confirm manual reset is required after fault clearance and that communications recovery cannot automatically restore motion.
- Prove hard overtravel and torque removal without a software round trip through either 7i84U.

5. Z-axis gravity and brake hazard

The HAL uses an asymmetric placeholder sequence: S-ON request rises immediately, brake release follows after 0.100 s; on disable the brake command drops immediately and S-ON is retained for 0.100 s. These are unmeasured commissioning placeholders, not proof that the brake is mechanically holding before torque disappears.

Hardware E-stop, mains loss, internal amplifier fault, or loss of amp supply can bypass the software delay. The cabinet requires a proven brake-drop-before-torque-loss path, measured under the worst approved load and bus condition. Until then, treat every hard stop or power loss as capable of dropping the head.

NO PERSON UNDER Z

Do not stand under the spindle head or rely on the brake during commissioning. Use an engineered mechanical support whenever the Z holding chain is not proven.

6. Retrofit-specific hazard register

Hazard	Consequence	Required control / evidence
Reversed analog polarity	Immediate runaway	D8 direction proof, zero-voltage test, one-axis first-move plan
Wrong resolver winding/phasing	Wrong position, discontinuity, oscillation	Ohmmeter pairs, exact suffix datasheet, scope traces, independent scale proof
Dual resolver excitation	Damage or invalid feedback	Continuity/isolation proof; 7i49 sole source
Wrong 7i84U TB1 landing	Supply short or card damage	Pins 1/2 VFIELDB; 3/4 VFIELDA; 5 VIN; 6/7/8 common
Bare P3 field wiring	7i80HDT FPGA damage	P3 empty; probe on opto-isolated 7i84U-B IN15
100 VAC on 7i84U output	Card damage, shock	Interposing relays and load-specific suppression
Smart-serial/Ethernet loss	Delayed or missing control	Remote watchdog, HostMot2 watchdog, latched inhibit, measured fault tests
ATC state error	Tool drop, collision, pinch injury	D13 hazard analysis, dry-cycle fixture, mutual exclusion, no automatic retry
Unverified FR-SX interface	Unexpected rotation or failed orient	Exact drive manual, terminal trace, measured handshake/timing

7. Live-power hold points

Milestone	Required deliverables
Control power	D1, D2 partial, D3, D5, D15
DC bus	Control-power set plus D6
Amplifier S-ON	Above plus D4, D7, D14
Z-brake release	Above plus measured Z row of D7
Spindle rotation	Above plus D12
First axis motion	Above plus D8 and D9
First automatic M6	Above plus D10, D11, D13